Suraksha-Route

AI-Powered Parametric Income Protection for Food Delivery Workers

Guidewire DEV Trails 2026

A modern weekly protection layer for gig workers whose income stops when the city does.

The Problem: Income Stops Instantly

Heavy Rain

Workers can't navigate safely

Flooding

Routes become impassable

Severe Pollution (AQI)

Health risk forces offline

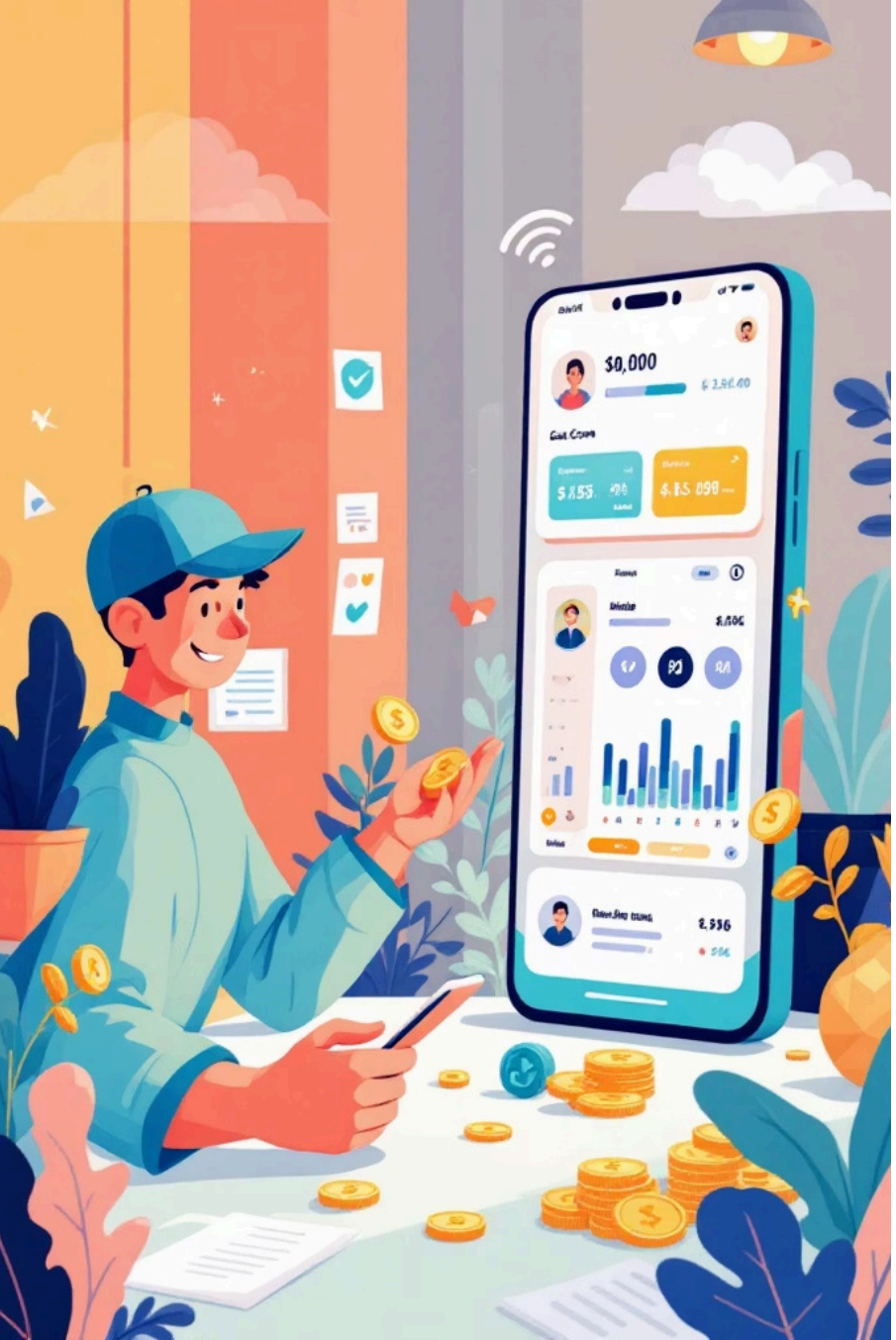
Heatwave

Extreme conditions halt work

App Outages

Platform unavailable

Traditional insurance is too slow and claim-heavy for daily, unpredictable income loss.



Our Solution: SurakshaRoute

01

Onboard

Worker signup & verification

02

Score

AI risk assessment by zone

03

Premium

Weekly pricing aligned to earnings

04

Activate

Policy goes live

05

Trigger

Disruption detected automatically

06

Payout

Instant claim settlement

Weekly pricing aligns with gig-worker cash flow; insurers gain speed and control.

Why Parametric, Why Food Delivery

Parametric insurance works best when the trigger is objective and the payout rule is pre-agreed. That means faster claims, less documentation, and a scalable operating model for frequent low-ticket protection.

Traditional claim insurance	SurakshaRoute parametric model
Manual loss proof required	Event-based triggers (objective)
Slow settlement (weeks)	Automatic claims (minutes)
High servicing cost	Low friction, scalable
Poor fit for short-duration loss	Built for gig-worker income cycles

Food delivery workers are hyper-local, weather-sensitive, and tightly linked to route availability and platform uptime.

AI/ML Modules and Their Role

Risk Scoring Model

Predicts disruption exposure using location, weather, timing, and income signals.

Dynamic Weekly Premium

Sets weekly pricing from risk, earnings, and hazard intensity.

Fraud Detection Engine

Flags spoofing, suspicious clusters, and mismatched route signals.

Trigger Monitoring Logic

Monitors weather, AQI, closures, curfews, and outage feeds.

Payout Estimation Engine

Calculates fair payouts from terms, duration, and earnings.

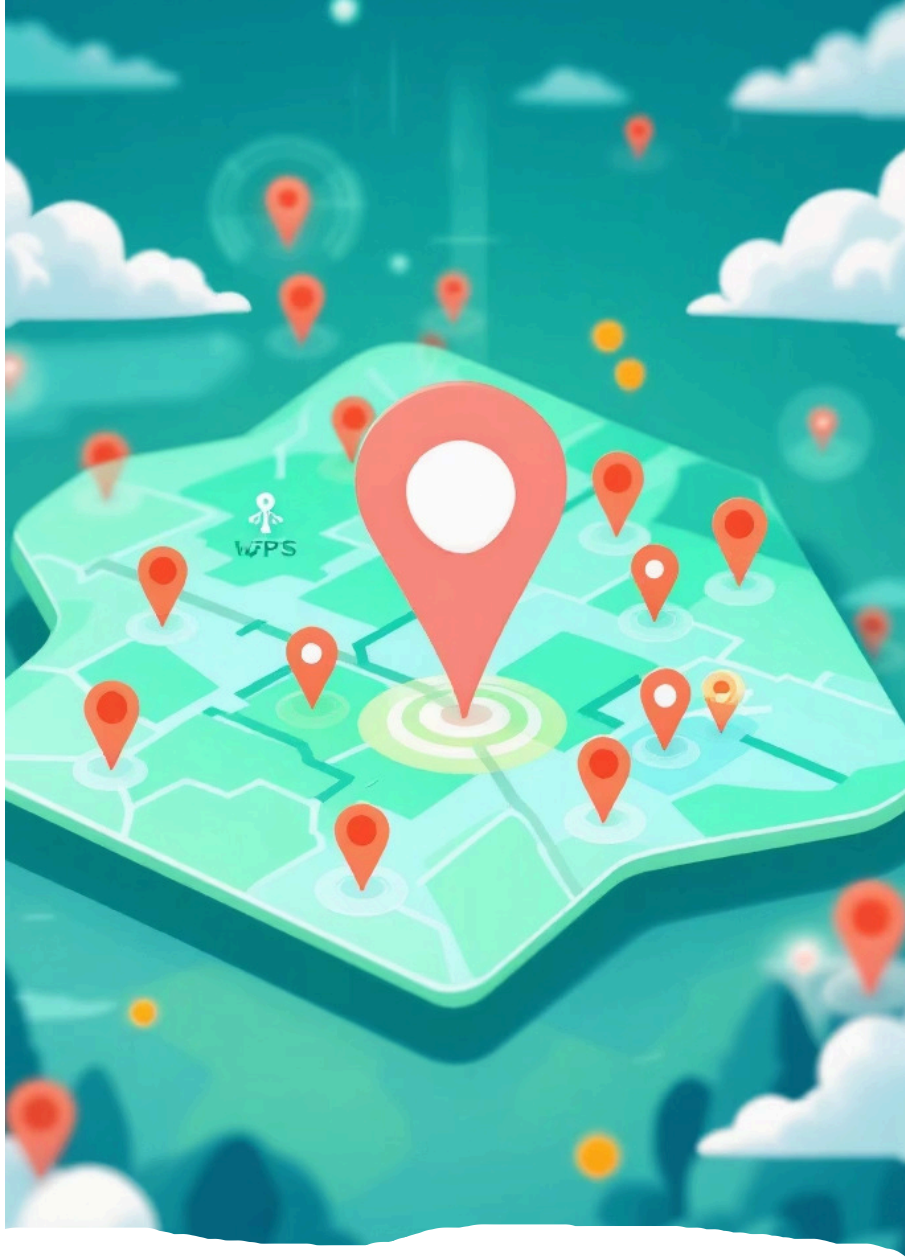
URGENT FRAUD RISK

GPS SPOOFING

New Challenge: GPS Spoofing Fraud

- Fraud ring uses GPS spoofing to fake location inside severe weather zones
- Triggers mass false payouts, threatening insurer liquidity
- Directly undermines trust in automatic claims
- GPS alone is never trusted in SurakshaRoute

Location is one signal among many—only meaningful when matched with route, device, timing, and event context.



Adversarial Defence + Fair UX

We separate genuine stranded workers from spoofing attackers using multi-signal verification and fraud-ring clustering. The goal is not to create friction for honest users — it is to make fraud expensive and honest claims fast.

Signal fusion

GPS, route, motion, device, network, and peer signals must all align.

Anomaly and cluster detection

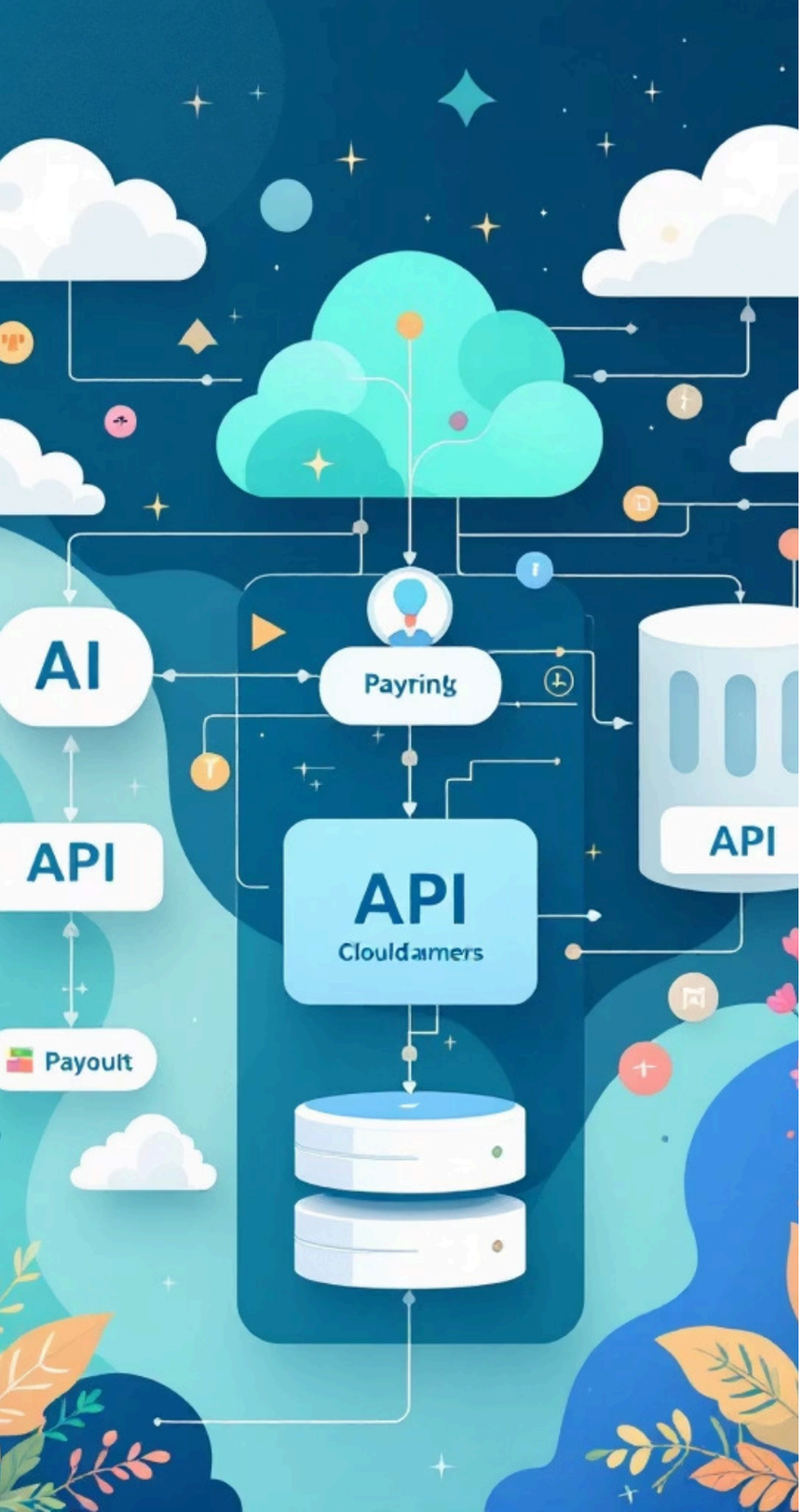
Spot reused devices, impossible travel, and coordinated timing across claim rings.

Fair decisioning

Auto-approve low risk, soft-hold medium risk, and review only high risk.

Worker-first resilience

Allow noisy field conditions, weak signals, and connectivity drops without unfair rejection.



System Architecture

Frontend

React/Next.js portal & dashboard.

Backend

FastAPI with modular AI/ML services.

Integrations

Weather, AQI, traffic, outage feeds.

Database

Policy, risk, trigger, claim, fraud events.

Public website

Product story and trust layer.

Worker portal

Onboarding, policy, premium, payouts.

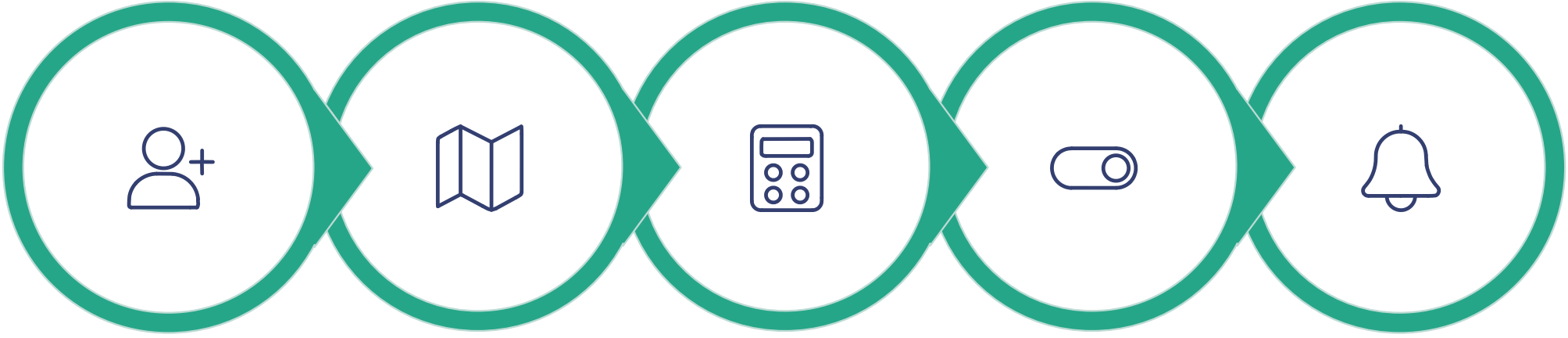
Admin dashboard

Risk, fraud, claims, controls.

AI/ML engine

Scoring, detection, triggers, estimates.

End-to-End Workflow



Signup **Zone Risk** **Weekly Premium** **Activate** **Detect Event**

The workflow is designed for speed and control: the worker gets a simple experience, while the platform continuously validates risk, detects disruption, and protects against manipulation before payout.

Impact, Demo Readiness, and Closing

Zero-Touch Claims

Instant payout, no manual review

Fraud Defence

Multi-signal verification blocks spoofing

Insurer Trust

Explainable triggers and controlled logic

Scalability

Extends across cities and gig personas

Demo-ready: working AI/ML engines, deployed dashboard, fraud defence, India-aligned architecture.

SurakshaRoute transforms disruption into fast, fair, and intelligent income protection.

Thank you. Q&A.